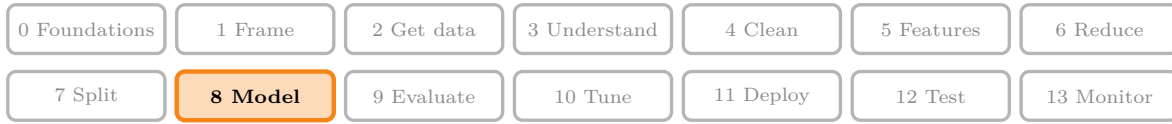


Naive Bayes: The Mathematics

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Where this fits: Pipeline map step 8 (Model). See the [Course map](#).



- **Builds on:** Classification problems ([Note ML-003](#)); Probability density function (PDF) ([Note ML-019](#)); Independent and mutually exclusive events ([Note MA-016](#)); Bayes' theorem ([Note MA-018](#)); Normal distribution ([Note MA-020](#)).

1. Overview

Key point: Starting from Bayes' theorem, the chain rule writes the joint probability exactly; the naive assumption (conditional independence) then simplifies it to $P(C_k) \times \prod P(x_i | C_k)$. The predicted class is the one that maximises this.

The previous Note used the Naive Bayes recipe on a small example. This Note derives it in general, step by step, and makes the “naive” assumption precise.

$$\begin{aligned}
 P(C_k | x) &\propto P(x_1, x_2, \dots, x_n, C_k) && \text{Bayes, evidence dropped} \\
 &= P(x_1 | x_2, \dots, x_n, C_k) P(x_2 | x_3, \dots, x_n, C_k) \cdots P(x_n | C_k) P(C_k) && \text{chain rule (exact)} \\
 &\approx P(x_1 | C_k) P(x_2 | C_k) \cdots P(x_n | C_k) P(C_k) && \text{naive assumption: drop the red parts} \\
 &= P(C_k) \prod_{i=1}^n P(x_i | C_k) && \text{the Naive Bayes formula} \\
 \hat{y} &= \arg \max_k P(C_k) \prod_{i=1}^n P(x_i | C_k) && \text{maximum a posteriori (MAP) rule}
 \end{aligned}$$

Figure 1: The derivation in five lines

2. Notation

Key point: Features $x = (x_1, \dots, x_n)$; K classes, C_1 to C_K for k up to K .

- An **observation** (G-1374; one record, a row of the data table) has n **features** (G-772; input variables, one column each), with values $x = (x_1, x_2, \dots, x_n)$. In the cricket example, $n = 3$: toss, venue and outlook.
- The **target** (G-1949; the output we predict) is one of K classes, $C_1, \dots, C_K$. In the cricket example, $K = 2$: win and loss.

The goal: for each class C_k , compute $P(C_k | x)$, then pick the largest.

	x_1	x_2	x_3	C_k
	toss	venue	outlook	result
new match	lost	Mumbai	sunny	?

features $x = (x_1, x_2, x_3)$, so $n = 3$
target: win or loss
 $K = 2$

Figure 2: The notation on the cricket example: three features and a target with two classes

Figure 2 puts the symbols on the cricket table: each feature column is one x_i , and the result column holds the class.

3. Step 1: Bayes' theorem without the evidence

Key point: $P(C_k | x)$ is proportional to $P(x | C_k) P(C_k)$ $P(C_k) = P(x_1, \dots, x_n, C_k)$.

We want the probability of each class given the features. Bayes' theorem turns that question round, into the probability of the features given the class, which the training data can count:

$$P(C_k | x) = \frac{P(x | C_k) P(C_k)}{P(x)}$$

The evidence $P(x)$ is the same for every class, so it does not affect which class is largest (the intuition Note). Dropping it, we write $\propto$, “is **proportional to**” (G-1584):

$$P(C_k | x) \propto P(x | C_k) P(C_k)$$

Figure 3 checks this on the cricket example of Section 6. The two scores are 0.040 for win and 0.056 for loss. Dividing both by their sum, the evidence 0.096, gives the posteriors 0.42 and 0.58: the bars change height but not order.

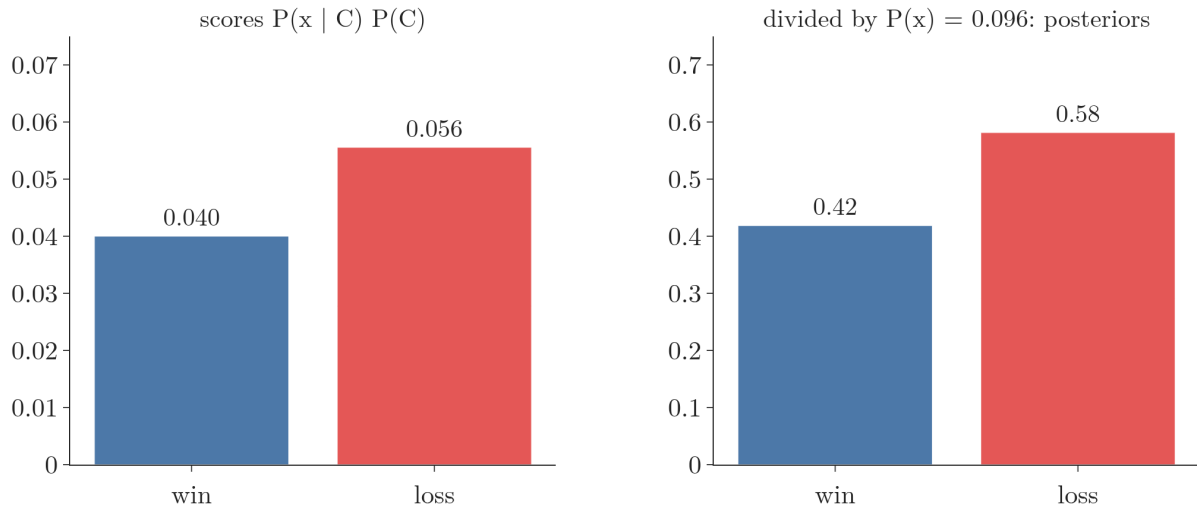


Figure 3: The cricket example. Left: the scores $P(x | C) P(C)$, 0.040 for win and 0.056 for loss. Right: the same scores divided by the evidence $P(x) = 0.096$, the posteriors 0.42 and 0.58. Both pick loss.

From the Bayes' theorem proof, $P(x | C_k)P(C_k) = P(x \cap C_k)$, the probability of the features and the class together. Writing the features out, with commas meaning “and”:

$$P(C_k | x) \propto P(x_1, x_2, \dots, x_n, C_k)$$

4. Step 2: the chain rule

Key point: Peel off one variable at a time with $P(A, B) = P(A | B) P(B)$. The rule is exact.

The probability of all the features and the class together is hard to count directly. The multiplication rule of the conditional probability Note, $P(A \cap B) = P(A | B) P(B)$, splits it into smaller pieces, one variable at a time. For two events: the chance of “toss lost and win” is the chance of a win times the chance of a lost toss given a win, $5/8 \times 1/5 = 1/8$, and indeed 1 of the 8 matches is a win after a lost toss.

Apply the same rule with $A = x_1$ and $B = (x_2, \dots, x_n, C_k)$:

$$P(x_1, \dots, x_n, C_k) = P(x_1 | x_2, \dots, x_n, C_k) P(x_2, \dots, x_n, C_k)$$

The last factor has the same form with one variable fewer, so apply the rule again, with $A = x_2$:

$$P(x_2, \dots, x_n, C_k) = P(x_2 | x_3, \dots, x_n, C_k) P(x_3, \dots, x_n, C_k)$$

Repeating until only x_n and C_k are left, and finally $P(x_n, C_k) = P(x_n | C_k) P(C_k)$:

$$P(x_1, \dots, x_n, C_k) = P(x_1 | x_2, \dots, x_n, C_k) P(x_2 | x_3, \dots, x_n, C_k) \cdots P(x_n | C_k) P(C_k)$$

This repeated splitting is the **chain rule of probability** (G-369). Nothing has been assumed yet: the chain rule is exact.

5. Step 3: the naive assumption

Key point: Assume each feature depends only on the class, not on the other features: $P(x_i | \text{other features}, C_k) = P(x_i | C_k)$.

The factors of the chain rule are hard to estimate. For the cricket match, $P(x_1 | x_2, x_3, C_k)$ is “the probability the toss was lost, given the venue was Mumbai, the weather was sunny and the match was won”. Few or no training observations match all those conditions, so the estimate is 0 or unreliable (the intuition Note).

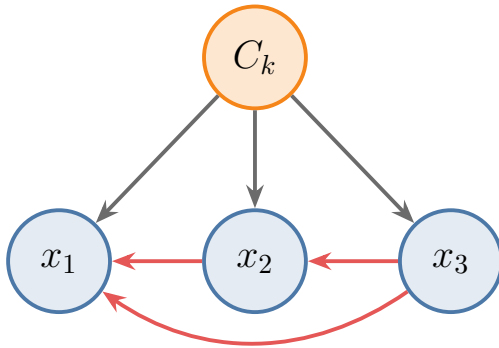
Naive Bayes assumes **conditional independence** (G-443): once the class is known, each feature is independent of the others. In symbols, for any features,

$$P(x_i | x_{i+1}, \dots, x_n, C_k) = P(x_i | C_k)$$

Conditional independence is the independence of the independent events Note, $P(A | B) = P(A)$, but holding **within each class**. Every chain-rule factor loses its other features (the red parts in Figure 1):

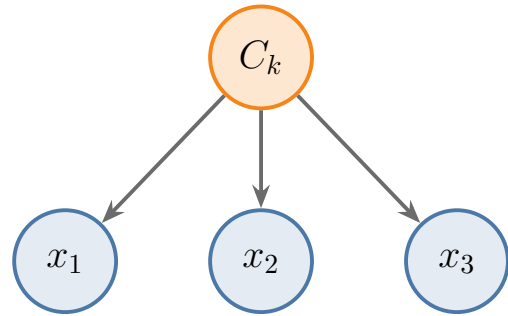
$$P(x_1, \dots, x_n, C_k) \approx P(x_1 | C_k) P(x_2 | C_k) \cdots P(x_n | C_k) P(C_k)$$

chain rule (exact)



features may depend on each other

naive assumption



each feature depends only on the class

Figure 4: Who depends on whom, for three features. Left: the chain rule allows every feature to depend on the class and on the other features (red arrows). Right: the naive assumption keeps only the arrows from the class

Figure 4 shows what the assumption removes: the red links between features. Each remaining factor needs only one feature and the class, which the training data can count reliably.

5.1 One factor, checked on the cricket data

Key point: The chain-rule factor rests on 1 match; the naive factor rests on 5.

Take the first factor for the class win, with the query of the intuition Note (toss lost, Mumbai, sunny).

1. **The chain-rule factor** is $P(\text{toss lost} \mid \text{Mumbai, sunny, win})$. Only the wins played in Mumbai in sunny weather count. There is 1 such match, and its toss was won. The estimate is $0/1 = 0$, from a single match.
2. **The naive factor** is $P(\text{toss lost} \mid \text{win})$. All 5 wins count, and the toss was lost in 1 of them. The estimate is $1/5 = 0.2$.

Chain-rule factor			Naive factor		
$P(\text{toss lost} \mid \text{Mumbai, sunny, win})$			$P(\text{toss lost} \mid \text{win})$		
<i>toss</i>	<i>venue</i>	<i>outlook</i>	<i>toss</i>	<i>venue</i>	<i>outlook</i>
won	Chennai	sunny	won	Chennai	sunny
won	Mumbai	sunny	won	Mumbai	sunny
lost	Chennai	sunny	lost	Chennai	sunny
won	Mumbai	overcast	won	Mumbai	overcast
won	Chennai	sunny	won	Chennai	sunny
uses 1 winning match toss lost in 0 of 1: estimate 0			uses all 5 winning matches toss lost in 1 of 5: estimate 0.2		

Figure 5: One factor on the five winning matches. Left: the chain-rule factor can use only the win played in Mumbai in sunny weather. Right: the naive factor uses all five wins.

Figure 5 shows the rows each estimate uses. The chain-rule factor is exact in principle, but one match cannot give a reliable estimate, and a 0 wipes out the whole product. The naive factor gives up exactness and gains data: every factor is counted over all the matches of the class.

6. The formula and the MAP rule

Key point: Score each class by $P(C_k) \prod P(x_i \mid C_k)$ and predict the largest: the maximum a posteriori rule.

Written with a product sign ($\prod$ is to multiplication what $\sum$ is to addition):

$$P(C_k \mid x) \propto P(C_k) \prod_{i=1}^n P(x_i \mid C_k)$$

To get actual probabilities, divide by the sum over all classes: $P(C_k \mid x) = \frac{1}{Z} P(C_k) \prod_i P(x_i \mid C_k)$, where Z is that sum, which equals the evidence $P(x)$.

The prediction is the class with the largest posterior:

$$\hat{y} = \arg \max_{k \in \{1, \dots, K\}} P(C_k) \prod_{i=1}^n P(x_i \mid C_k)$$

($\arg \max$ (G-210) means “the k that gives the maximum”.) This is the **maximum a posteriori (MAP) rule** (G-1157). On the cricket example, it gives 0.040 for win and 0.056

for loss, so $\hat{y} = \text{loss}$.

Figure 6 builds the two scores one factor at a time, for a match where the toss was lost, the venue was Mumbai and the weather sunny. Win starts ahead on its prior, 5/8 against 3/8. The second factor turns it round: only 1 of the 5 wins came after a lost toss, against 2 of the 3 losses. From there on loss stays ahead.

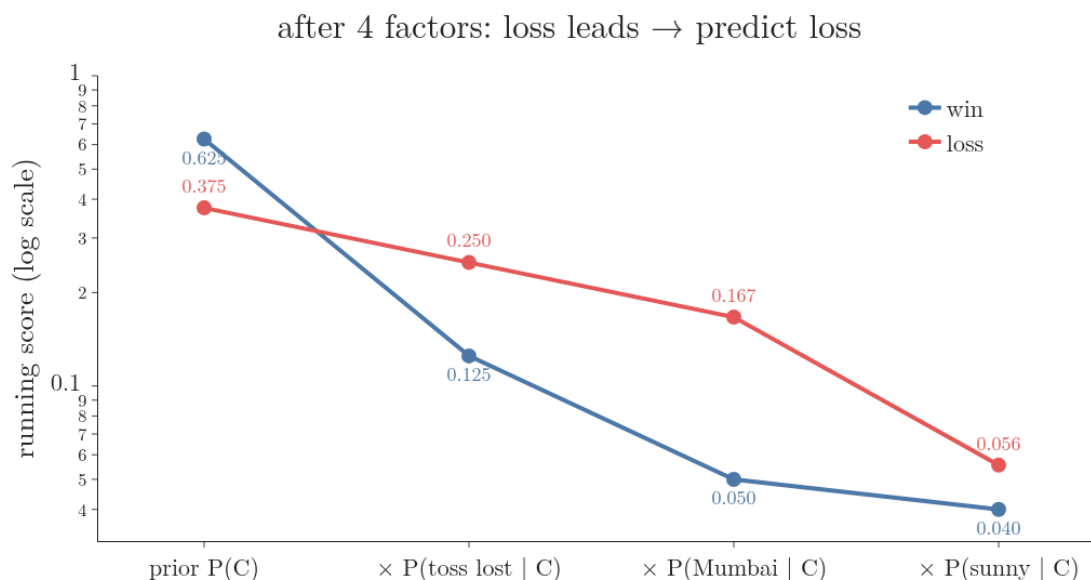


Figure 6: The MAP rule on the cricket example, one factor per frame: the prior, then the probability of each observed feature given the class. Win leads on the prior (0.625 against 0.375) but falls behind after the toss factor; the final scores are 0.040 and 0.056, so the prediction is loss. Log scale.

7. When the assumption fails

Key point: The chain rule is exact; the naive product is only exact when the features really are independent within each class. Strongly related features get counted twice.

A simulation with two binary features and 200,000 observations shows the difference:

Features within a class	Exact $P(x_1, x_2 C)$	Chain rule	Naive product
independent	0.476	0.476	0.477
x_2 is a copy of x_1	0.800	0.800	0.639

When the features are independent, the naive product matches. When x_2 simply repeats x_1 , the naive product multiplies the same evidence twice and is wrong.

An everyday picture: two friends tell us the same rumour, but both read it in the same newspaper. The rumour feels twice as certain, yet we have only one source.

The same information counted again and again makes Naive Bayes over-confident

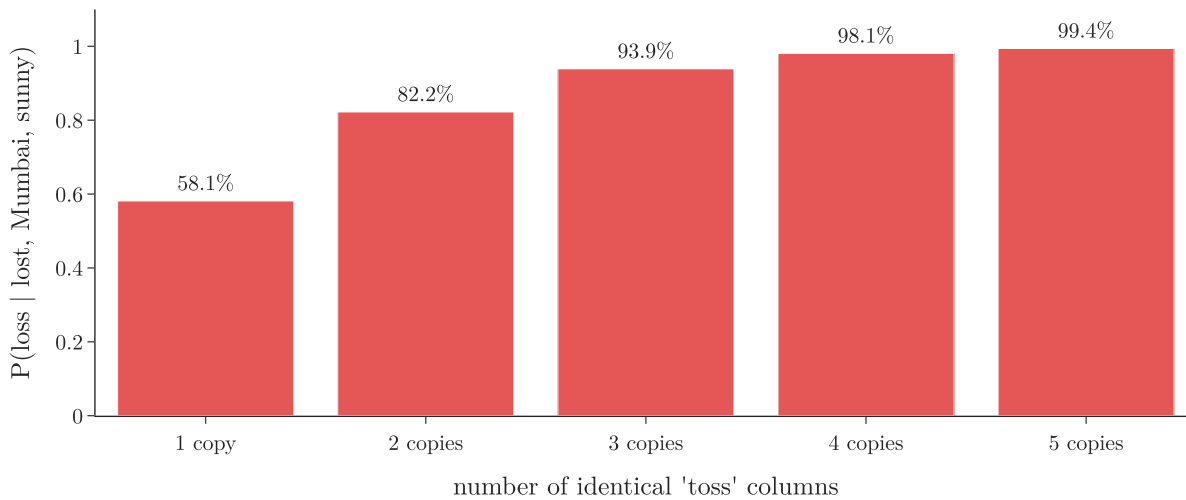


Figure 7: Copying a feature makes Naive Bayes over-confident

Figure 7 shows the effect on the cricket prediction: each extra copy of the “toss” feature pushes $P(\text{loss})$ further, from 58% with one copy to 99.4% with five, although no new information was added.

In practice, Naive Bayes often still picks the right class even when its independence assumption is wrong, because only the **order** of the scores matters for the prediction. Its probabilities, however, tend to be too extreme, so they should not be trusted as exact (Domingos and Pazzani 1997; scikit-learn user guide §1.9).

Extra: Multiplying many probabilities, each below 1, produces very small numbers (the log loss Note). Implementations therefore add logarithms instead: $\log P(C_k) + \sum_i \log P(x_i \mid C_k)$. The largest sum identifies the same class.

8. Summary

Step	Result
Bayes, drop the evidence	$P(C_k x) \propto P(x_1, \dots, x_n, C_k)$
Chain rule (exact)	$\prod_i P(x_i x_{i+1}, \dots, x_n, C_k) \times P(C_k)$
Naive assumption	$P(x_i \dots, C_k) = P(x_i C_k)$
Formula	$P(C_k) \prod_i P(x_i C_k)$
MAP rule	predict $\arg \max_k$ of the formula

9. Sources

Built from

- CampusX, “Naive Bayes Classifier | Part 7 | Mathematics behind Naive Bayes Algorithm”, YouTube, <https://www.youtube.com/watch?v=2PVRG45eVrY>

Other references

- **Domingos and Pazzani 1997:** Domingos, P. and Pazzani, M. “On the Optimality of the Simple Bayesian Classifier under Zero-One Loss.” *Machine Learning* 29, 103–130, 1997.
- **scikit-learn user guide:** Section 1.9, “Naive Bayes”, scikit-learn 1.9.

10. Key terms

Term	Meaning
Chain rule of probability	Writing a joint probability as a product of conditional probabilities, one variable at a time
Conditional independence	Independence that holds once a third variable (here the class) is known
Proportional to ($\propto$)	Equal up to a constant factor that is the same for every class
$\arg \max$	The value of the variable that makes an expression largest
MAP rule	Maximum a posteriori: predict the class with the largest posterior probability